International Journal of Computer Vision (2026) 134:501
https://doi.org/10.1007/s11263-026-03071-z

DogMo: A Large-Scale Multi-View RGB-D Dataset for 4D Canine Motion Recovery

Zan Wang[1] · Siyu Chen[1] · Luya Mo[1] · Xinfeng Gao[1] · Yuxin Shen[1] · Lebin Ding[1] · Wei Liang[1,2,3]

Received: 9 December 2025 / Accepted: 23 September 2026
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Abstract

We present ***DogMo***, a large-scale multi-view RGB-D video dataset capturing diverse canine movements for real canine motion recovery. ***DogMo*** comprises 1.2k motion sequences collected from 10 unique dog instances, covering 11 action types and over 220 minutes of recordings. Each motion is captured by five synchronized RGB-D cameras, providing multi-view RGB-D observations, masks, keypoints, and action labels. Compared with existing dog datasets, ***DogMo*** offers rich variation in both motion and breed, provides synchronized multi-view RGB-D observations at a larger scale, and supports systematic evaluation for real canine motion recovery. Leveraging ***DogMo***, we establish four motion recovery benchmark settings that support evaluation across monocular and multi-view, RGB and RGB-D inputs. We also provide a three-stage, instance-specific fitting pipeline, which produces observation-consistent SMAL-based canine motion annotations by progressively refining dog body shape and pose through coarse alignment, dense correspondence supervision, and temporal regularization. We hope ***DogMo*** can facilitate future research on animal motion recovery, multi-view reconstruction, and animal behavior modeling.

Keywords Canine motion recovery · Multi-view RGB-D video

1 Introduction

Over the past decade, human body recovery from images and videos has made remarkable progress, driven by the release of numerous large-scale public datasets (Mahmood et al., 2019; Black et al., 2023), standardized benchmarks (Ionescu et al., 2013; Marcard et al., 2018), and advanced algorithmic frameworks (Bogo et al., 2016; Kanazawa et al., 2018; Li et al., 2022; Shin et al., 2024). As interest in animal welfare and the pet industry continues to grow, efforts have emerged to extend human motion recovery methods to animals (Biggs, Boyne, Charles, Fitzgibbon, & Cipolla, 2020; Rüegg, Zuffi, Schindler, & Black, 2022; Sabathier, Mitra, & Novotny, 2024; Zuffi, Kanazawa, Jacobs, & Black, 2017) for behavior analysis and virtual asset creation. However, progress in this area remains limited, primarily due to the lack of comprehensive animal datasets, which constrains advancements in algorithm design and evaluation.

Acquiring animal motion data is inherently inefficient and costly due to the lack of a tailored toolchain and the fact that animals, unlike humans, cannot follow instructions

Communicated by Dima Damen.

Zan Wang, Siyu Chen, Luya Mo and Xinfeng Gao have contributed equally to this work.

✉ Wei Liang
liangwei@bit.edu.cn

Zan Wang
wangzan@bit.edu.cn

Siyu Chen
3120240925@bit.edu.cn

Luya Mo
3120230882@bit.edu.cn

Xinfeng Gao
3220231219@bit.edu.cn

Yuxin Shen
shenyuxin@bit.edu.cn

Lebin Ding
lebinding749@gmail.com

[1] School of Computer Science & Technology, Beijing Institute of Technology, Beijing 100081, China

[2] Yangtze Delta Region Academy of Beijing Institute of Technology, Jiaxing, Jiaxing 314003, China

[3] State Key Laboratory of General Artificial Intelligence, BIGAI, Beijing 100081, China

Published online: 28 October 2026



Fig. 1 We present ***DogMo***, a large-scale multi-view RGB-D video dataset capturing diverse canine movements using 5 synchronized low-cost cameras. ***DogMo*** contains 1.2*k* motion sequences from 10 unique dog instances, spanning 11 action types and totaling over 220 minutes of recordings

consistently during capture. Consequently, publicly available animal datasets are scarce and suffer from two major issues:

- *Lack of synchronized multi-view RGB-D observations*. Most existing datasets (Biggs et al., 2019; Yu et al., 2021; Yang et al., 2022) consist of monocular RGB videos and lack synchronized multi-view recordings with depth observations, limiting the study of multi-view and RGB-D settings for real canine motion recovery.
- *Limited scale and diversity*. Existing datasets (Kearney et al., 2020; Luo et al., 2022; Shooter et al., 2024a, b) typically contain less than one hour of footage, with minimal variation in motion types and animal instances.

In this work, we focus on motion recovery for dogs, one of the most common companion animals in human life. To address the aforementioned issues, we introduce ***DogMo***, a large-scale multi-view RGB-D video dataset capturing dog movements, collected using 5 low-cost RGB-D cameras that are carefully synchronized and calibrated, as illustrated in Fig. 1. These cameras are evenly distributed around a capture zone to record dog movements from multiple viewpoints. Within this zone, dogs are encouraged to perform natural behaviors such as running, jumping, and interacting with humans. In total, ***DogMo*** collects 1.2*k* motion sequences from 10 unique dog instances, with a total duration exceeding 220 minutes. We emphasize that ***DogMo*** is designed as a practical RGB-D observation dataset and benchmark, rather than a marker-based MoCap or high-fidelity 4D-scan dataset. Built upon this dataset, we establish four canine motion recovery settings for fitting and evaluating SMAL-based canine motion from (1) monocular RGB videos, (2) multi-view RGB videos, (3) monocular RGB-D videos, and (4) multi-view RGB-D videos.

Despite significant progress in human motion recovery, estimating dog motion from visual observations presents unique challenges. First, the wide variation in body size across breeds limits the model's generalization capability. Yet prior works (Zuffi et al., 2017; Biggs et al., 2020; Yang et al., 2023) primarily address monocular shape estimation, which inherently suffers from ambiguities in recovering absolute body shape. Second, dogs' fur-covered and often textureless appearance makes body parts visually indistinct, hindering accurate localization. Approaches such as Rüegg et al., (2022), Sabathier et al. (2024) depend on pre-trained keypoint detectors or part segmentation models, but they overlook the potential of temporal motion priors to regularize parts and joints across time.

To address these challenges, we first extend D-SMAL (Rüegg et al., 2022; Zuffi et al., 2017) with an additional scaling factor to better handle the large variation in dog body shapes. Building on this extension, we provide a three-stage instance-specific SMAL fitting pipeline that fits the model parameters to input observations in a

coarse-to-fine manner. The first stage performs lightweight coarse alignment to initialize the global translation, orientation, and body shape, reducing the optimization space and improving fitting stability. In the second stage, following Sabathier et al. (2024), we leverage Continuous Surface Embedding (CSE) (Neverova et al., 2020) for dense correspondence supervision, together with regularization terms for leg-crossing, shape, and pose constraints. Finally, the third stage refines the motion sequence with a temporal smoothness loss to improve temporal stability and reduce frame-to-frame jitter caused by noisy per-frame estimates.

We summarize the contributions as: (1) We introduce ***DogMo***, a large-scale multi-view RGB-D dataset for real canine motion recovery, providing synchronized RGB-D observations, masks, keypoints, action labels, and fitted SMAL-based motion annotations. (2) We establish four benchmark settings spanning monocular/multi-view and RGB/RGB-D inputs, supporting systematic evaluation of real canine motion recovery methods. (3) We provide a three-stage instance-specific SMAL fitting pipeline that produces observation-consistent canine motion annotations through coarse alignment, dense correspondence supervision, and temporal refinement. (4) We conduct extensive experiments across different input settings, baselines, ablations, and metric analyses to evaluate the proposed benchmark and fitting pipeline.

2 Related Work

2.1 Model-Free 3D Animal Reconstruction

Model-free 3D animal reconstruction from images or videos either employs instance-specific 3D shape optimization (Cashman & Fitzgibbon, 2012; Yang et al., 2021; Wu et al., 2022; Yang et al., 2023) or trains neural networks to directly regress 3D structures without requiring 3D supervision (Wu et al., 2023b; Liu et al., 2023b; Wu et al., 2023a; Yao et al., 2023a; Aygun & Mac Aodha, 2024; Li et al., 2024). Both approaches avoid reliance on *parametric models* and primarily leverage differentiable rendering to enforce 2D supervision, such as silhouettes, keypoints, and semantic correspondences. While the former is typically tailored to individual instances, the latter aims to generalize across diverse articulated animal species by learning from large-scale internet image collections. More recent efforts have shifted toward building *neural implicit representations* for animals (Luo et al., 2022; Yang et al., 2022; Sinha et al., 2023; Tan et al., 2023; Tu et al., 2025; Yang et al., 2025), or distilling 3D reconstructions from 2D generative models (Ho et al., 2020; Rombach et al., 2022) by extending Score Distillation Sampling (SDS) (Poole et al., 2023) to image-conditioned multi-view generation models (Liu et al., 2023c; Qian et al., 2024; Shi et al., 2024). Our focus is on model-based 4D dog motion recovery from a wide range of visual inputs, spanning monocular and multi-view, RGB and RGB-D inputs.

2.2 Model-Based Animal Pose and Shape Estimation

Model-based animal pose estimation from 2D observations, such as 2D (Cao et al., 2019; Yu et al., 2021; Yang et al., 2022; Zeng et al., 2024; Ye et al., 2024) and 3D (Kearney et al., 2020; Bala et al., 2020; Muramatsu et al., 2022; Sosa & Hogg, 2023; Shooter et al., 2024a) keypoint detection, plays a crucial role in animal behavior analysis. Leveraging parametric animal models like SMAL (Zuffi et al., 2017) and ABM (Badger et al., 2020), recent approaches aim to directly estimate pose and shape parameters from images, as these models provide strong priors over animal body shape and articulation. These methods typically either optimize (Zuffi et al., 2017; Wang et al., 2021; Li et al., 2021; Sabathier et al., 2024) or regress (Biggs et al., 2020; C. Li & Lee, 2021; C. Li, Yang, et al., 2024b; Rüegg, Tripathi, Schindler, Black, & Zuffi, 2023; Rüegg et al., 2022) the parameters from observations, often relying on 2D re-projection losses due to the absence of ground-truth 3D annotations. Some studies (Zuffi et al., 2019; Lyu et al., 2025; Shooter et al., 2024a) instead train models to directly regress 3D pose and shape using synthetic data or motion capture datasets, though these approaches are often limited in data diversity and scale. We provide a practical three-stage pipeline for progressively fitting the SMAL model to video sequences, demonstrating its robust performance across various visual inputs.

2.3 Animal-Related Datasets

Early datasets for animal analysis mainly comprise 2D images or videos annotated with 2D labels, enabling tasks such as classification (Khosla et al., 2011), detection (Liu et al., 2023a), segmentation (Mu et al., 2020; Deane et al., 2024), 2D/3D pose estimation (Biggs et al., 2019; Cao et al., 2019; Biggs et al., 2020; Yu et al., 2021; Joska et al., 2021; Yang et al., 2022; Yao et al., 2023b), and behavior understanding (Li et al., 2020; Ng et al., 2022; Ma et al., 2023). Compared to these datasets, some works leverage simulators to synthesize animal images (Zuffi et al., 2019; Shooter et al., 2024a, b; Lyu et al., 2025), which provide access to 3D pose. Other efforts rely on expensive capture devices, such as motion capture systems or 3D scanners, to acquire high-quality 3D data of horses (Li et al., 2024a; Zuffi et al., 2024), birds (Naik et al., 2023), and rats (Marshall et al., 2021). Kearney et al. (2020) collect RGB-D data of dogs using a Kinect V2 camera, supplemented with MoCap to obtain accurate 3D poses. Xu et al. (2023) generate pseudo ground-truth 3D annotations by manually fitting the SMAL

model to 2D images. Rodent3D (Patel et al., 2023) provides multi-view, multi-modal video recordings of mice to facilitate pose tracking and behavioral analysis. The emergence of Neural Radiance Field (NeRF) has inspired datasets (Luo et al., 2022; Sinha et al., 2023) that leverage multi-view imaging or video to reconstruct detailed 3D neural animals. This work introduces a new large-scale multi-view RGB-D video dataset capturing diverse canine movements, aimed at facilitating research in dog motion recovery and related fields.

3 *DogMo* Dataset

3.1 Data Capture

Device Setup. We collect dog motion data using a low-cost capture system equipped with 5 ORBBEC Astra 2[1] RGB-D cameras. The cameras are evenly distributed along a circular area with a radius of approximately 2.5 meters, all oriented toward the center of the capture zone. Each camera is positioned at a height of about 0.45 meters, with a random pitch angle within ± 10 degrees. Figure 2 exhibits the overview of the capture setup.

For camera calibration, we leverage the OpenCV library (Bradski, 2000) to manually calibrate each pair of adjacent cameras, thus deriving the relative poses of any two cameras. To synchronize all cameras, we employ the Sync Hub Dev[2] device provided by ORBBEC.

Capture Procedure. During data collection, a dog trainer guides the dog to perform specific actions within the capture zone by issuing semantic commands, such as running, jumping, or playing with humans. We ensure that the dog remains within the capture area so that at least three cameras can simultaneously capture the dog's full body. Since the trainer needs to actively guide the dog, occasional occlusions of the dog by the trainer are unavoidable from certain viewpoints.

3.2 Dataset Statistics

In ***DogMo***, each camera records synchronized RGB and depth streams, with RGB images at a resolution of 1920×1080 and depth images at 1600×1200. The RGB and depth frames are hardware-aligned within the camera. The system operates at 15 FPS. ***DogMo*** contains $1.2k$ motion clips collected from 10 unique dog instances, as displayed in Fig. 2b. Each motion clip lasts between 0.87 and 19.07 seconds, covering 11 action types ranging from running to playing with humans. In total, the dataset provides over 220 minutes of motion footage, amounting to more than $1M$ frames across all camera views. To further support research, we annotate each frame with binary masks and keypoints of the dog, and assign a semantic label to each motion. We provide an overview of ***DogMo*** in Fig. 3, and a comparison with existing related datasets is summarized in Table 1.

In addition to the main indoor benchmark, we further collect a small complementary outdoor subset to increase the environmental diversity of ***DogMo***. This subset contains 219 motion clips from 3 dog instances, covering actions from the same action vocabulary as the indoor data. In total, it provides 23 minutes of motion footage, amounting to 103,470 frames across all camera views. The outdoor data are captured in a fixed outdoor scene under different illumination conditions, including daytime and nighttime recordings. Compared with the controlled indoor setup, this subset introduces additional visual variations such as natural lighting changes, outdoor backgrounds, and low-light observations. We use the outdoor subset as a complementary set for preliminary robustness and generalization analysis, while keeping the original indoor data as the main benchmark due to its larger scale and more controlled capture setting.

4 Method

4.1 Problem Formulation

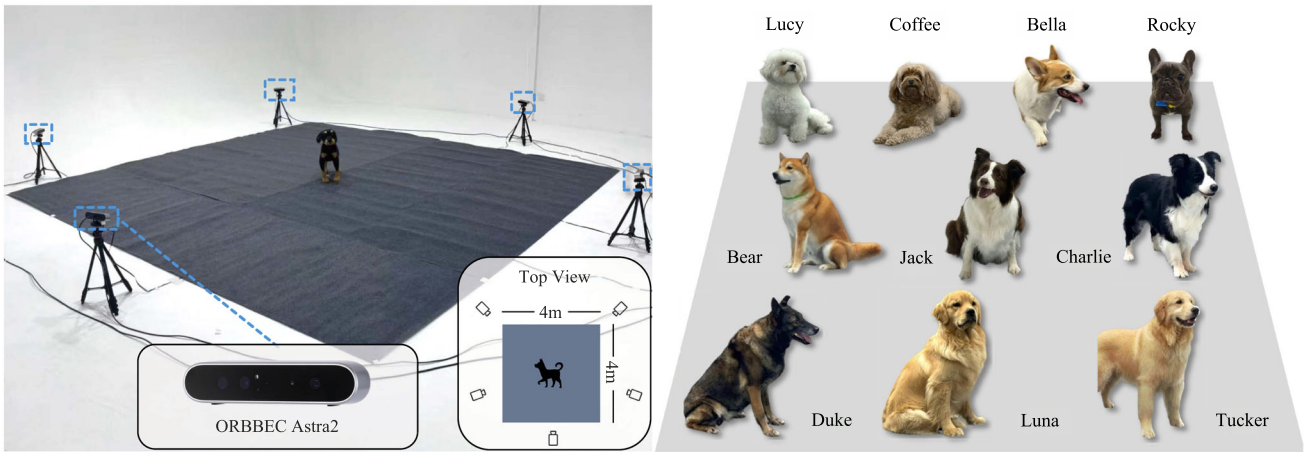
Given a video sequence $\{\mathbf{I}_t\}_{t=1}^{T}$, our goal is to reconstruct the 3D posed mesh of a canine subject, parameterized by D-SMAL (Zuffi et al., 2017; Rüegg et al., 2023) parameters $\{\mathbf{S}_t\}_{t=1}^{T}$. Each frame $\mathbf{I}_t$ can be a single-view RGB, multi-view RGB, single-view RGB-D, or multi-view RGB-D input, where T denotes the number of frames. For simplicity, we denote the RGB and depth images as C_t and D_t, respectively. The D-SMAL parameters at each frame are represented as a 5-tuple $\mathbf{S}_t = (\beta, \theta_t, \gamma_t, \phi_t, s)$, where $\beta \in \mathbb{R}^{30}$ represents the shape parameters, $\theta_t \in \mathbb{R}^{6N}$ denotes the N joint rotations in 6D representation, $\gamma_t \in \mathbb{R}^3$ is the global translation, and $\phi_t \in \mathbb{R}^6$ is the global orientation. To accommodate the broader range of dog sizes, we introduce s as an additional scaling parameter for resizing the body mesh, extending the original D-SMAL formulation. Given these parameters, the D-SMAL model finally outputs a posed 3D mesh, defined as

$$\mathcal{V}, \mathcal{F} = \mathcal{M}(\beta, \theta, \phi) \times s + \gamma, \tag{1}$$

where $\mathcal{V} \in \mathbb{R}^{3889\times 3}$ are the mesh vertices, $\mathcal{F} \in \mathbb{N}^{7774\times 3}$ is a list of triangular faces, and $\mathcal{M}(\cdot)$ is the original skinning function of D-SMAL.

[1] ORBBEC Astra 2: https://www.orbbec.com/products/structured-light-camera/astra-2/

[2] Sync Solutions: https://www.orbbec.com/products/camera-accessories/sync-solutions/



(a) Capture System (b) Dogs

Fig. 2 Motion Capture. We employ 5 synchronized RGB-D cameras to record 10 individual dogs

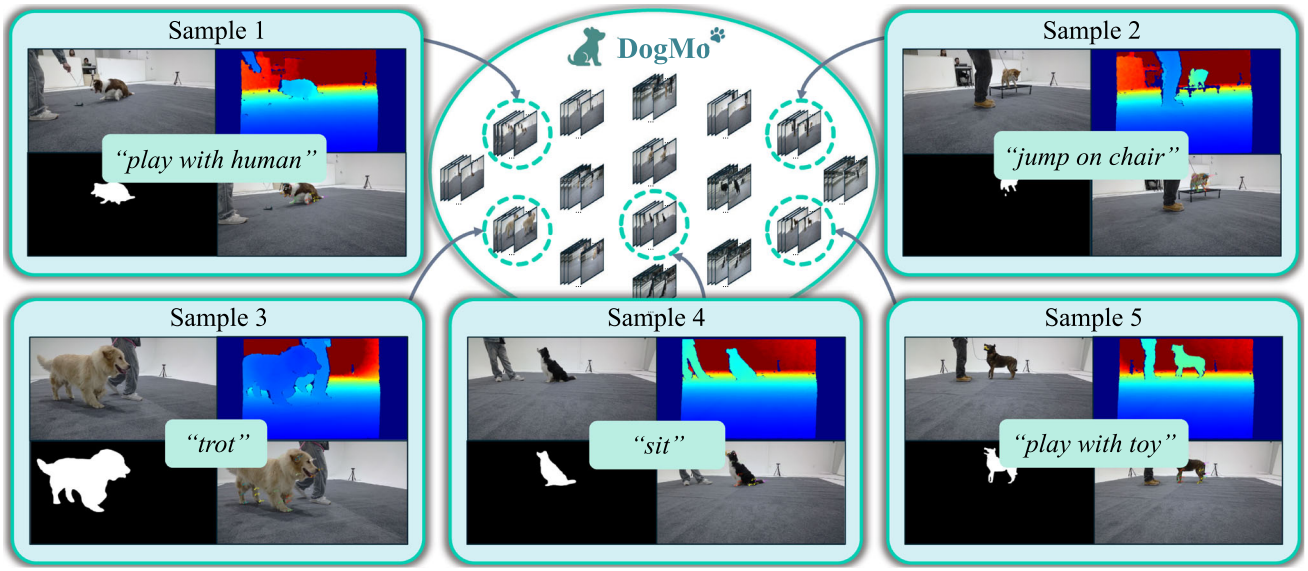


Fig. 3 Overview of ***DogMo***. ***DogMo*** provides five-view RGB-D image sequences for each motion recording, along with corresponding masks, keypoints, and semantic (*i.e.*, action label) annotations

Table 1 Comparison between ***DogMo*** and existing related datasets.

Dataset	Modality	FPS	Resolution	#Views	#Frames	#Subjects	#Actions	Real/Synth
CoP3D (Sinha et al., 2023)	RGB	–	720 × 1280	1	600*k*	4200	1*	Real
Sydog-video (Shooter et al., 2024b)	RGB	25	1024 × 1024	1	87.5*k*	5	5	Synth
Digidogs (Shooter et al., 2024a)	RGB-D	–	1280 × 720	1	27.9*k*	8	12	Synth
RGBD-Dogs (Kearney et al., 2020)	RGB-D	6	2560 × 1440	6	1.9*k*	7	5	Real
DogMo (Ours)	RGB-D	15	1920 × 1080	5	1*M*	10	11	Real

The notation * indicates that nearly all dog motions in the CoP3D dataset are in a sprawling posture throughout the entire video.

4.2 Canine Motion Recovery

To encourage temporal smoothness of the results, we follow Sabathier et al. (2024) and model the time-varying tuple $(\theta_t, \gamma_t, \phi_t)$ as the output of a multilayer perceptron (MLP) network conditioned on a smooth timestep embedding $\tau(t)$, instead of directly optimizing the D-SMAL parameters. Here, we represent

$$\theta_t, \gamma_t, \phi_t = \text{MLP}_\psi(\tau(t)). \tag{2}$$

In contrast, the body shape parameters β and the scaling factor s are treated as time-invariant and are directly optimized. During the optimization, we adopt a three-stage optimization pipeline that progressively recovers the MLP parameters ψ, body shape parameter β, and the scaling factor s, from coarse to fine, as detailed below. Figure 4 presents the overview of the optimization pipeline.

4.2.1 Stage 1: Coarse Alignment

In the first stage, we initialize the parameters (γ_t, ϕ_t, s) by coarsely aligning the projected 3D model with the observed data. This coarse alignment is guided by three frame-specific loss terms defined below.

Chamfer Mask Loss. The Chamfer Mask Loss encourages alignment between the projected D-SMAL mesh and the observed dog silhouette, defined as:

$$\mathcal{L}_t^{\text{mask}} = \text{CD}\big(\{u \mid u \in C_t \odot M_t\}, \{\mathcal{R}(X) \mid X \in (\mathcal{V}_t, \mathcal{F}_t)\}\big). \tag{3}$$

Here, u denotes the 2D pixels occupied by the dog, derived from the segmentation mask M_t; and M_t is predicted by the off-the-shelf Grounded SAM 2 (Ren et al., 2024) using the text prompt "dog." $\mathcal{R}(\cdot)$ denotes the camera projection operation, which projects 3D points onto the 2D image plane using a fixed projection matrix shared across all frames under the assumption of a static camera setup. X represents 3D points sampled from the posed mesh $(\mathcal{V}_t, \mathcal{F}_t)$. $\text{CD}(\cdot, \cdot)$ computes the Chamfer distance between two point sets. To better capture leg motion, we increase the sampling density in regions corresponding to legs, improving their silhouette alignment.

Sparse Keypoint Loss. To further enforce alignment between the keypoints of the 3D model and observed 2D keypoints, we apply the Sparse Keypoint Loss defined as:

$$\mathcal{L}_t^{\text{kp}} = \|h(C_t) - \mathcal{R}(\mathcal{J}_t)\|, \tag{4}$$

where $h(C_t)$ denotes the 2D keypoints detected from the image C_t using the sparse dog keypoint detector from BARC Rüegg et al., 2022, and $\mathcal{R}(\mathcal{J}_t)$ represents the projected corresponding 3D keypoints $\mathcal{J}_t$ from the estimated D-SMAL mesh.

Chamfer Depth Loss. To exploit per-pixel depth supervision when depth input is available, we incorporate a Chamfer Depth Loss defined as:

$$\mathcal{L}_t^{\text{depth}} = \text{CD}\big(\{v \mid v \in \mathcal{R}^{-1}(M_t \odot D_t)\}, \{X \mid X \in (\mathcal{V}_t, \mathcal{F}_t)\}\big). \tag{5}$$

Here, $\mathcal{R}^{-1}$ denotes the inverse camera projection operation, which uses the depth map and the camera projection matrix to lift the masked depth image pixels into 3D space. This loss measures the Chamfer Distance between the resulting point cloud and the sampled points on the estimated dog mesh. We use this term in settings that include depth input.

4.2.2 Stage 2: Joint Optimization

In the second stage, we optimize all parameters using a more comprehensive, frame-specific loss function that includes three additional loss terms beyond those previously introduced.

CSE Keypoint Loss. Following Sabathier et al. (2024), we incorporate a CSE Keypoint Loss to enforce denser correspondence supervision. This loss is defined as:

$$\mathcal{L}_t^{\text{cse}} = \sum_{u \in M_t \odot C_t} \|u - \mathcal{R}(\text{CSE}(\mathcal{V}, u)\|, \tag{6}$$

which penalizes the distance between each foreground pixel u in the dog mask and the 2D projection of its corresponding 3D vertex on the recovered D-SMAL mesh. The correspondence is provided by the image-to-CSE predictor $\text{CSE}(\cdot)$ (Neverova et al., 2020).

Leg Cross Loss. To prevent unnatural intersections between the left and right legs of the posed D-SMAL model, we introduce a Leg Cross Loss defined as:

$$\mathcal{L}_t^{\text{cross}} = \sum_{(J^l, J^r) \in \mathcal{J}_t^{\text{foot}}} \mathbb{1}_{\geq \delta} \cdot \exp(-\|(J^l - J^r)\|, \tag{7}$$

which penalizes overly close joint pairs from opposite legs to ensure physically plausible and natural leg motion. Here, $\mathcal{J}_t^{foot}$ denotes a predefined set of joint pairs corresponding to the left and right legs, and $\mathbb{1}_{\geq \delta}$ is an indicator function that outputs 1 when $\exp(-\|(J^l - J^r)\|$ exceeds threshold δ, and 0 otherwise.

Shape & Pose Prior Loss. To impose structural and kinematic constraints on the deformed mesh, we follow Rüegg et al. (2022), Zuffi et al. (2017) and apply a prior loss defined as:

$$\mathcal{L}^{\text{prior}} = \mathcal{L}^{\text{shape}} + \sum_{t=1}^{T} \mathcal{L}_t^{\text{pose}}. \tag{8}$$

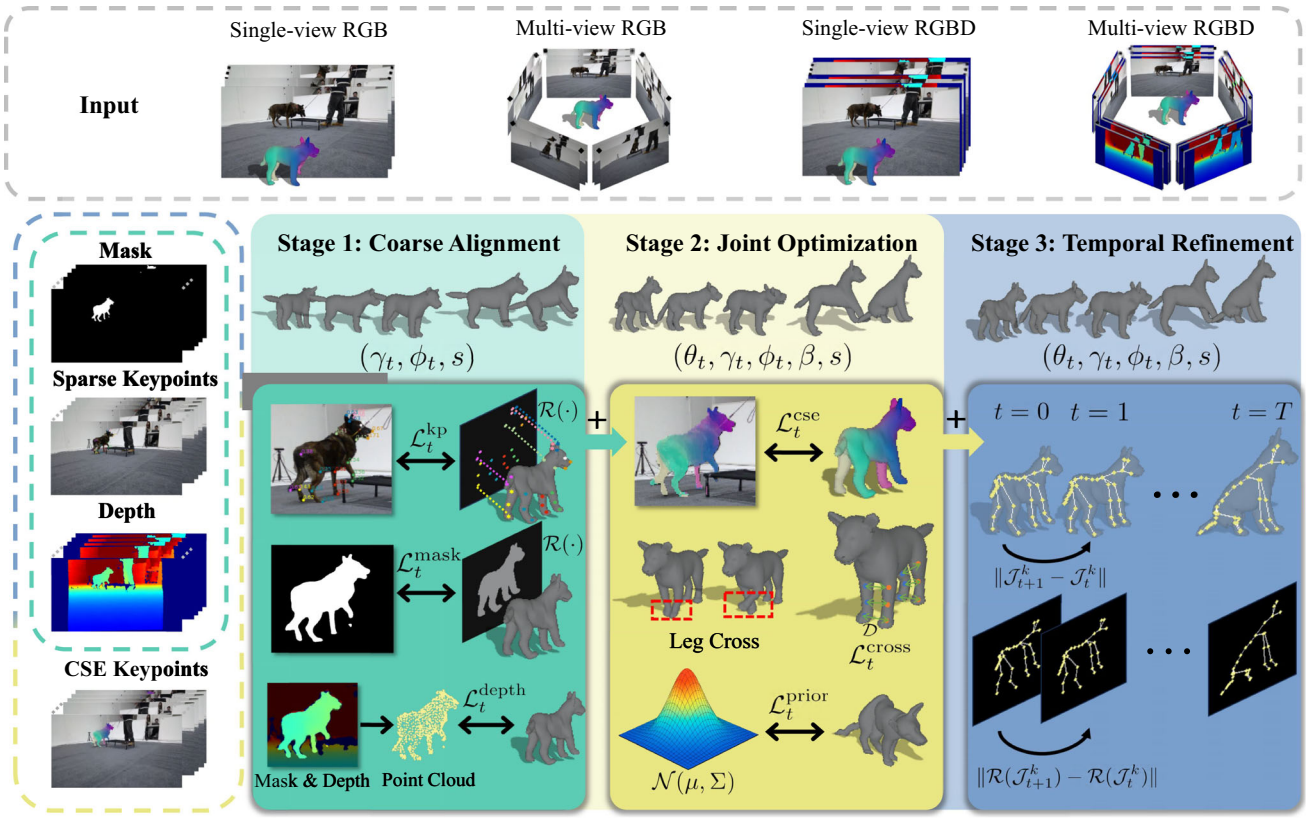


Fig. 4 Optimization pipeline. Our pipeline includes three stages, progressively refining the dog's body shape and pose through coarse alignment, dense correspondence supervision, and temporal regularization

The shape prior term $\mathcal{L}^{\text{shape}}$ encourages the D-SMAL shape coefficients β to match a prior distribution. The pose prior term $\mathcal{L}_t^{\text{pose}}$ constrains the estimated D-SMAL pose parameters θ_t to remain within plausible limits by applying a Gaussian prior penalty. Details are presented in *Sup. Mat.*.

4.2.3 Stage 3: Temporal Refinement

While the previous loss terms are computed independently for each frame, they do not enforce temporal consistency across the sequence. The third stage incorporates a Temporal Smoothness Loss, in addition to the above comprehensive frame-wise losses, to further refine the motion sequence:

$$\mathcal{L}^{\text{temp}} = \sum_{t=1}^{T-1} \sum_{k=1}^{|\mathcal{J}|} \big(\|\mathcal{J}_{t+1}^k - \mathcal{J}_t^k\| + \|\mathcal{R}(\mathcal{J}_{t+1}^k) - \mathcal{R}(\mathcal{J}_t^k)\| \big), \tag{9}$$

which penalizes abrupt changes in both 3D joint positions and their 2D projections.

This stage is mainly designed to improve temporal smoothness rather than per-frame reconstruction accuracy. It suppresses frame-to-frame fluctuations by penalizing abrupt changes in both 3D joint positions and their 2D projections. We note that temporal smoothing may introduce a slight trade-off with frame-wise alignment, since it can marginally affect per-frame metrics such as IoU and F-score. In Sec. 5, we empirically show that Stage 3 improves temporal stability while largely preserving reconstruction quality.

4.3 Implementation Details

Method Usage Across Different Settings. In the first and second stages, the loss is computed by averaging over mini-batches of frames sampled from video sequences. In the third stage, motion is further refined by optimizing over a sampled continuous sequence segment. Based on ***DogMo***, we establish four evaluation setups for motion recovery. For setups involving multi-view inputs, we average the loss across all views. The key difference between setups with and without depth input lies in using Chamfer Depth Loss, which is applied only when depth data is available.

Hyperparameter Configurations. In our implementation, we employ two separate MLPs to map the time-varying global translation and orientation, as well as body pose, *i.e.*, MLP_{TR} and MLP_θ. MLP_{TR} consists of 3 layers with 16 hidden units per layer, while MLP_θ has 3 layers with 64 hidden units

per layer. The number of optimization steps is the product of a specified multiplier and the frame number. For single-view sequences, we set the multipliers as 5, 20, and 5 for the first, second, and third stages, respectively; for multi-view sequences, the corresponding multipliers are 10, 25, and 5. We perform optimization using the Adam optimizer. All experiments are implemented in PyTorch and run on a single NVIDIA GeForce RTX 4090 GPU. Refer to *Sup. Mat.* for more details.

5 Experiments

5.1 Evaluation Setup

We evaluate methods under four settings: single-view RGB, multi-view RGB, single-view RGB-D, and multi-view RGB-D. Single-view settings use one camera view as input, while multi-view settings use synchronized observations from all five views and aggregate the losses across views. RGB settings use masks, keypoints, and RGB-based correspondences, while RGB-D settings additionally use depth observations through the Chamfer Depth Loss.

Metrics. We evaluate the methods using six metrics: (1) *IoU*, the Intersection-over-Union between the rendered silhouette of the estimated SMAL model and the ground-truth dog segmentation; (2) IoU_{w5}, the average IoU over the worst 5% of frames in each sequence, reflecting the method's robustness; (3) *F-score* (Wang et al., 2018), which quantifies the distance between the estimated dog mesh and the inverse-projected dog point cloud from RGB-D images; (4) $Pene_{\%}$, the proportion of mesh vertices that penetrate the floor over a sequence; (5) *Jitter* (m), measuring temporal smoothness; and (6) *Foot Skating (FS)* (m), which evaluates the naturalness of foot contact by capturing sliding artifacts. Since F-score measures geometric consistency between the recovered mesh and the observed RGB-D point cloud, we use it as an important evaluation metric for RGB-D observation consistency. The computation details of these metrics are presented in *Sup. Mat.*.

Baselines

To the best of our knowledge, no open-source baselines are directly available for dog motion recovery from multi-view or RGB-D videos.

To strengthen the benchmark with learning-based methods, we include two recent animal pose and shape estimation baselines: BITE Rüegg et al., (2023) and AniMer (Lyu et al., 2025). BITE estimates dog pose and shape from monocular RGB images. We use its officially released model and pre-trained weights, and evaluate it under the single-view RGB setting. AniMer is a family-aware Transformer-based method for animal pose and shape estimation. Since AniMer supports supervised training on dataset-specific annotations, we train it on ***DogMo*** and evaluate it under all four recovery settings, including single-view RGB, multi-view RGB, single-view RGB-D, and multi-view RGB-D.

We also compare with two optimization-based methods: SMALify (Biggs et al., 2020; Biggs, 2020) and AnimalAvatar (Sabathier et al., 2024). SMALify fits the SMAL model to monocular observations through optimization, and we evaluate it under the single-view RGB setting using the officially released implementation and configuration. AnimalAvatar is originally designed for recovering an animatable dog model from single-view RGB video sequences. We further adapt it to handle RGB-D input by incorporating depth supervision, and use the default model configuration and hyperparameters.

For the multi-view settings, we additionally implement a classical triangulation-based baseline, denoted as Multi-View SMAL-Fit. Specifically, we first detect 2D dog keypoints in each view and triangulate them into 3D keypoints using the calibrated multi-view camera parameters. We then fit the SMAL model by minimizing the distance between the SMAL joints and the triangulated 3D keypoints, together with the silhouette/depth alignment terms when available and the standard shape and pose priors.

We also include SAM 3D (Chen et al., 2026) as an additional qualitative reference. Since SAM 3D reconstructs a mesh from a single RGB frame but does not recover metric scale, articulated skeletons, or temporally consistent motion parameters, it is not directly comparable under our 4D canine motion recovery metrics. We therefore only provide qualitative comparisons.

Except for AnimalAvatar (Sabathier et al., 2024), we emphasize that the compared methods are originally formulated for single-frame pose and shape estimation rather than video-based motion recovery. We therefore apply them frame by frame to each video sequence and concatenate the per-frame reconstructions in temporal order. This protocol preserves their native inference setting, while highlighting the difference between single-frame reconstruction methods and video-level motion recovery methods.

In addition, we include four ablated variants of our pipeline, w/o stage 1, w/o stage 3, w/o leg-cross, and w/o scale, to assess the contribution of coarse alignment initialization, temporal refinement, leg-cross regularization, and the additional global scaling factor s, respectively.

All implementation details of the baselines and ablated variants are provided in the *Sup. Mat.*.

5.2 Results

Quantitative Results

We present the quantitative results in Table 2, from which we draw the following several key observations.

Table 2 Quantitative results

Method	Single-view RGB						Single-view RGB-D					
	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓
BITE	0.5971	0.3077	0.3490	0.4420	0.1422	0.0573	-	-	-	-	-	-
AniMer	0.5983	0.3721	0.3113	0.4480	0.1909	0.0549	0.5900	0.3675	0.7041	0.2807	0.1886	0.0571
SMALify	0.6133	0.4058	0.2324	0.4363	0.0895	0.0435	-	-	-	-	-	-
AnimalAvatar	0.6116	0.5197	0.0283	**0.2702**	0.0298	**0.0144**	0.6036	0.4905	0.4748	0.2039	0.0260	0.0184
w/o leg-cross (Ours)	0.6448	0.5226	0.3575	0.3996	0.0219	0.0397	0.6351	0.4901	0.6619	0.6024	0.4842	0.3360
w/o scale (Ours)	0.6139	0.5072	0.1860	0.2253	0.0217	0.0355	0.6284	0.5190	0.6169	0.2734	0.0186	0.0209
w/o stage 1 (Ours)	0.6024	0.4842	0.3360	-	-	-	0.6074	0.4901	0.6619	-	-	-
w/o stage 3 (Ours)	0.6491	0.5235	0.3576	0.3698	0.0211	0.0366	**0.6424**	**0.5235**	0.7666	**0.1261**	0.0140	0.0167
Full (Ours)	**0.6505**	**0.5247**	**0.3593**	0.3653	**0.0206**	0.0361	0.6414	0.5228	**0.7673**	0.1278	**0.0137**	**0.0166**
Method	Multi-view RGB						Multi-view RGB-D					
	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓
AniMer	0.5725	0.3618	0.8575	0.2339	0.1426	0.0568	0.5809	0.3943	0.8662	0.2342	0.1423	0.0563
Multi-View SMAL-fit	0.6121	0.4334	0.9043	0.0762	0.0143	0.0334	0.6118	0.4357	0.9057	0.0756	0.0145	0.0331
w/o leg-cross (Ours)	0.6217	0.4242	0.9086	0.0727	0.0138	0.0331	0.6282	0.4442	0.9078	0.0707	0.0134	0.0331
w/o scale (Ours)	0.6194	0.4295	0.9021	0.0884	0.0128	0.0196	0.6206	0.4349	0.9032	0.0726	0.0128	0.0188
w/o stage 1 (Ours)	0.6084	0.4358	0.9096	-	-	-	0.6122	0.4403	0.9134	-	-	-
w/o stage 3 (Ours)	**0.6271**	**0.4496**	**0.9107**	0.0700	0.0124	**0.0188**	**0.6300**	**0.4557**	**0.9137**	0.0699	0.0123	0.0188
Full (Ours)	0.6258	0.4490	**0.9107**	**0.0688**	**0.0120**	**0.0188**	0.6286	0.4549	**0.9137**	**0.0687**	**0.0119**	**0.0187**

Bold values indicate the best results for each evaluation metric within the corresponding setting

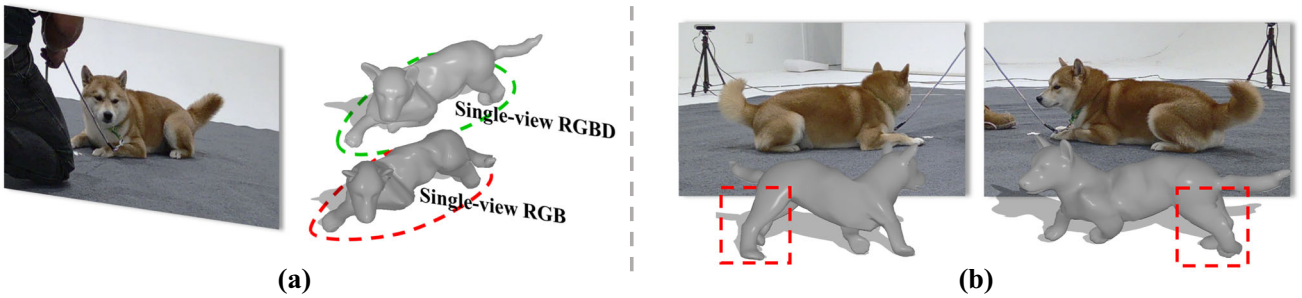


Fig. 5 Ambiguity in dog mesh recovery. **a** Without multi-view RGB-D observations, the optimization method using only single-view RGB input fails to recover the dog's absolute position and size accurately. **b** The resulting mesh also exhibits noticeable artifacts when viewed from other perspectives

Table 3 Quantitative results on outdoor subset

Time of Day	Single-view RGB						Single-view RGB-D					
	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓
Daytime	0.6489	0.5480	0.3247	-	0.0425	-	-	-	-	-	-	-
Nighttime	0.6049	0.4955	0.3863	0.2747	0.0473	0.0499	0.6020	0.4997	0.6906	0.2268	0.0323	0.0452
Time of Day	Multi-view RGB						Multi-view RGB-D					
	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓	IoU ↑	IoU_{w5} ↑	F-score ↑	$Pene_{\%}$ ↓	Jitter ↓	FS ↓
Daytime	0.6243	0.5086	0.8075	-	0.0302	-	-	-	-	-	-	-
Nighttime	0.5793	0.3965	0.8291	0.0760	0.0342	0.0500	0.5838	0.3977	0.8337	0.0661	0.0401	0.0676

Table 4 Optimization efficiency

Method	Setting	Stage 1	Stage 2	Stage 3	Total	Setting	Stage 1	Stage 2	Stage 3	Total
AnimalAvatar	Single-view RGB	-	-	-	114.68	Single-view RGB-D	-	-	-	117.47
Ours	Single-view RGB	2.75	12.70	3.33	18.78	Single-view RGB-D	2.83	13.02	3.40	19.25
Ours	Multi-view RGB	19.82	53.12	11.22	84.16	Multi-view RGB-D	20.38	54.52	11.45	86.35

We compute the average running time (in minutes) of optimization required by each model. Lower is better

- Although the single-view RGB setting achieves a high IoU on the optimized view, it performs much worse in F-score, $Pene_{\%}$, Jitter, and Foot Skating. This is because IoU is a 2D silhouette metric and can overestimate reconstruction quality when evaluated on the same view used for optimization. To verify this, we project the single-view RGB results to all five views: the IoU drops from 0.6505 to 0.2799, while multi-view RGB achieves an average IoU of 0.6258. This shows that the high single-view IoU mainly reflects view-specific silhouette fitting, whereas multi-view and depth inputs provide stronger 3D constraints. These observations are consistent with the qualitative results in Fig. 5.
- Our method significantly outperforms SMALify (Biggs et al., 2020; Biggs, 2020) and AnimalAvatar (Sabathier et al., 2024) across IoU and F-score, demonstrating that these baselines struggle to accurately fit dog silhouettes and recover global 3D positions and body sizes. Although AnimalAvatar uses a similar optimization strategy, its reliance on the original D-SMAL model limits its ability to handle large variations in dog body sizes. Its one-stage, frame-independent optimization also lacks proper initialization and temporal constraints, often yielding suboptimal results. While AnimalAvatar yields better $Pene_{\%}$ and Foot Skating, this is primarily because it produces meshes that float above the floor. *Such non-contact meshes naturally result in zero penetration and foot skating, which spuriously improve these metrics.*
- The first-stage coarse alignment is crucial for motion recovery, particularly in the single-view RGB-D setting, as it provides favorable initialization and prevents suboptimal convergence in the subsequent dense optimization. We also observe that varying the initial value of s greatly affects the results of the model *w/o stage 1*, especially for $Pene_{\%}$, Jitter, and FS; therefore, we omit these results in Table 2. Additional analysis is provided in Sect. 5.3. The third-stage temporal refinement further improves the final results in the single-view RGB setting, and its

impact diminishes in multi-view or depth-rich scenarios, where the input provides sufficient information for accurate recovery.

In the setting of motion recovery from single-view RGB video, the absence of multi-view RGB-D observations introduces ambiguity in the reconstructed dog mesh, *i.e.*, the estimated results cannot reliably determine the true position and size of the dog. As shown in Fig. 5, we compare results from the same recording using single-view RGB and single-view RGB-D inputs. The reconstruction from single-view RGB shows clear deviations in body position and size compared to that using inputs with 3D cues, which aligns with the F-score results reported in Table 2. Besides, the resulting mesh from single-view RGB input also exhibits noticeable artifacts when viewed from other perspectives.

Results on Outdoor Subset. We further evaluate our full model on the outdoor subset to study its robustness under less controlled environments. We split the outdoor data into daytime and nighttime subsets according to the illumination condition, and report the results separately in Table 3. Compared with the indoor benchmark, the performance on the outdoor subset is noticeably lower. This suggests that outdoor canine motion recovery remains more challenging due to the more complex scene layout, less clean background, occlusion from surrounding vegetation, and illumination variations such as shadows. We also observe a clear performance gap between daytime and nighttime recordings. The nighttime subset yields worse results than the daytime subset, mainly because low-light observations make silhouette, keypoint, and dense correspondence estimation less reliable. We do not evaluate RGB-D settings for daytime outdoor recordings because the infrared-based depth sensing of ORBBEC Astra 2 becomes unreliable under sunlight interference. RGB-D results are therefore reported only for nighttime recordings, where the depth observations are more reliable.

Qualitative Results. Fig. 6 shows the qualitative results of our method across the four setups. Our method effectively recovers dog motion from various input types, ranging from single-view to multi-view and from RGB to RGB-D inputs.

Figure 7 further presents qualitative results on the outdoor subset. Despite more complex backgrounds, grass occlusion, shadows, and varying illumination, our method can recover plausible canine motions from observations. Meanwhile, the outdoor examples also reveal remaining challenges, as the results become more sensitive to imperfect masks and keypoints, especially under nighttime illumination.

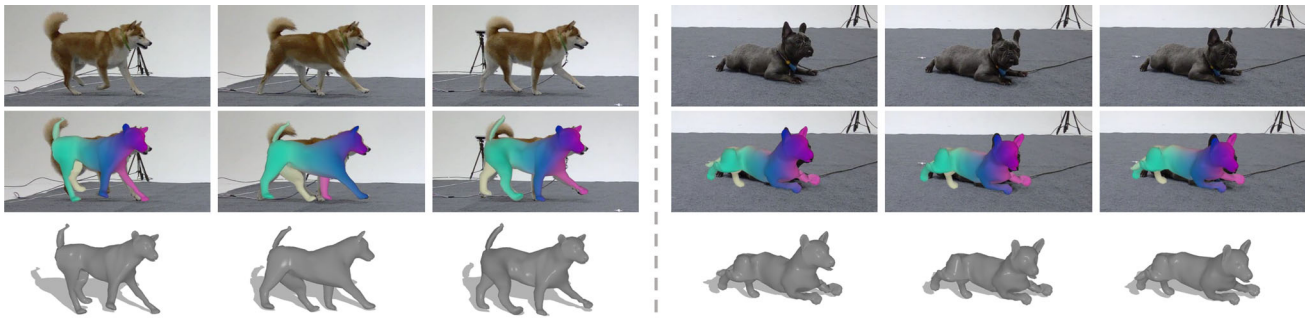
Qualitative Comparison. In Fig. 8, we compare the qualitative results of the baseline models with those of our full model. When recovering dog motion from a single-view RGB image, the comparison shows that both SMALify (Biggs et al., 2020; Biggs, 2020) and AnimalAvatar (Sabathier et al., 2024) fail to capture the dynamics of body parts, particularly the leg movements, resulting primarily in global translation and orientation changes. Without the initialization provided by the first stage, the method struggles to reconstruct the complete motion.

We also provide video demonstrations in the supplementary demo video.

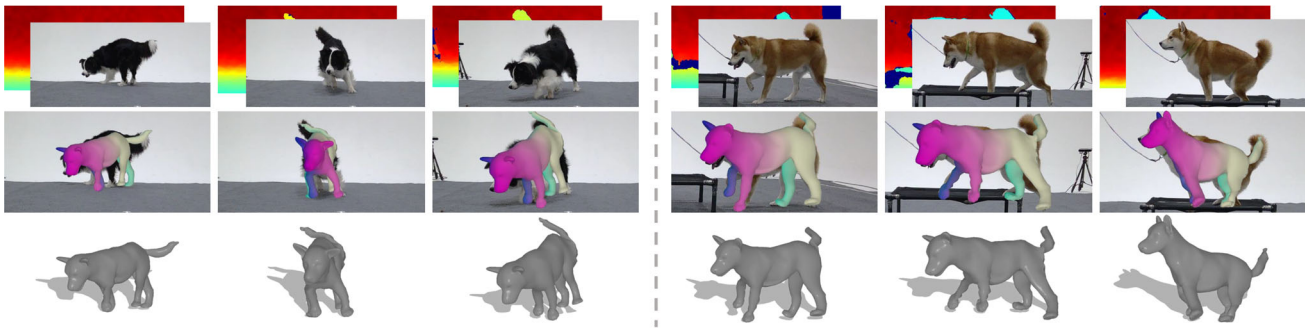
Generalization. We emphasize that although our method adopts an instance-specific optimization pipeline, it generalizes well across diverse cases. This generalization stems from its reliance on dog masks and keypoints predicted by pretrained models, *i.e.*, Grounded SAM 2 (Ren et al., 2024), which are known for strong robustness across varied scenarios. In addition, the carefully designed cost functions are broadly applicable, as they are independent of specific pixel values and remain effective across different instances. To further demonstrate this, we evaluate our method on CoP3D (Sinha et al., 2023), which contains in-the-wild videos, and DigiDogs (Shooter et al., 2024a), which provides synthetic RGB-D dog data. Since neither dataset provides multi-view imaging, we conduct only single-view evaluations: RGB setting on CoP3D and RGB-D setting on DigiDogs, respectively. Qualitative results are shown in Fig. 9. For quantitative results, our method achieves IoU 0.7190 and IoU_{w5} 0.5912 on CoP3D, and IoU 0.6644, IoU_{w5} 0.5832, and F-score 0.7946 on DigiDogs, which are slightly higher than those on ***DogMo***. We attribute this improvement to two factors: (1) the dog motions in CoP3D are predominantly sprawling with limited variation, making them easier to optimize; and (2) DigiDogs is a synthetic dataset with minimal observation noise, enabling more accurate reconstruction.

Optimization Efficiency. To evaluate optimization efficiency, we report the runtime of our model for each stage and the entire pipeline, as shown in Table 4. Most computation is spent in the second stage, which involves all optimization terms and requires more iterations. The cost further increases with multi-view observations. Notably, our method converges in fewer iterations than AnimalAvatar (Sabathier et al., 2024), resulting in shorter overall runtime. This improvement stems from the strong initialization in the first stage and the use of a scaling parameter that resizes the body mesh, effectively reducing the optimization space in the second stage.

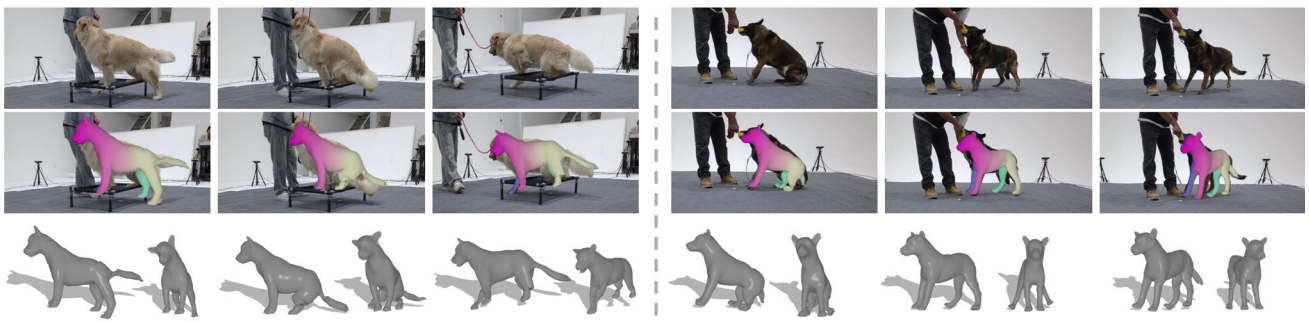
Failure Cases. We present several failure cases produced by our method in Fig. 10. As shown in the results, when the dog is curled up in a resting position, it becomes difficult to distinguish individual body parts due to the uniform texture across the whole body, leading to inaccurate reconstruction. Additionally, when the dog is occluded, particularly the distinguishable body parts such as the head, the limited visual information also leads to poor reconstruction of the body shape.



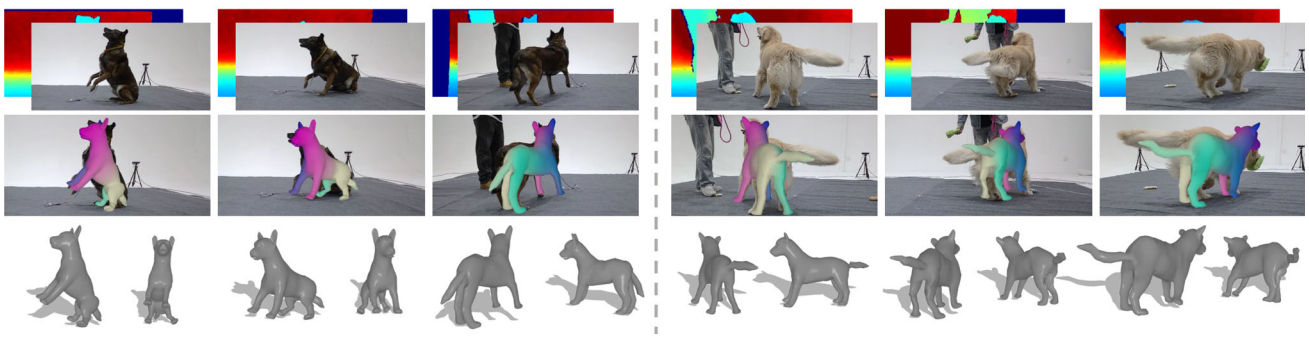
(a) Optimization Results on Single-view RGB Setting



(b) Optimization Results on Single-view RGB-D Setting

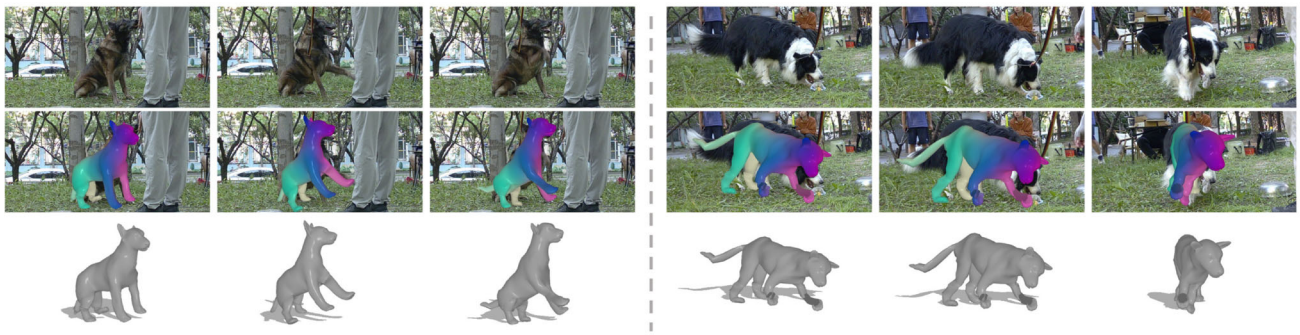


(c) Optimization Results on Multi-view RGB Setting

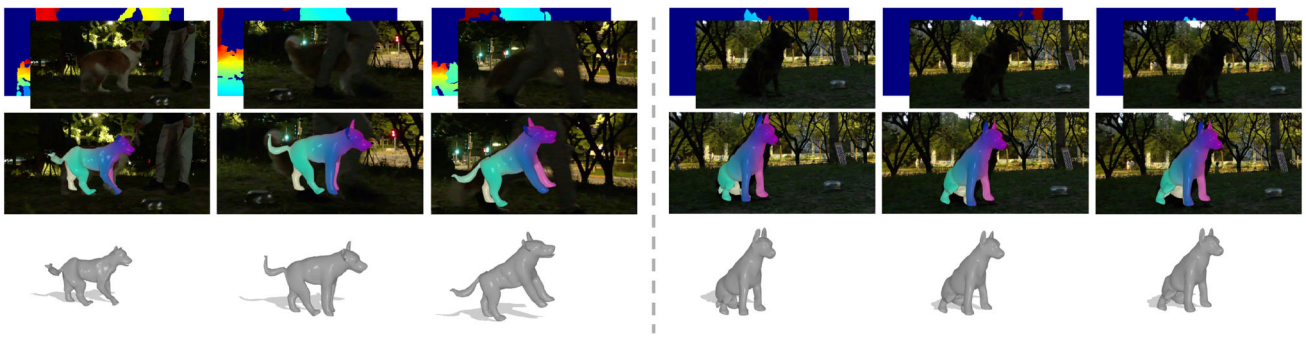


(d) Optimization Results on Multi-view RGB-D Setting

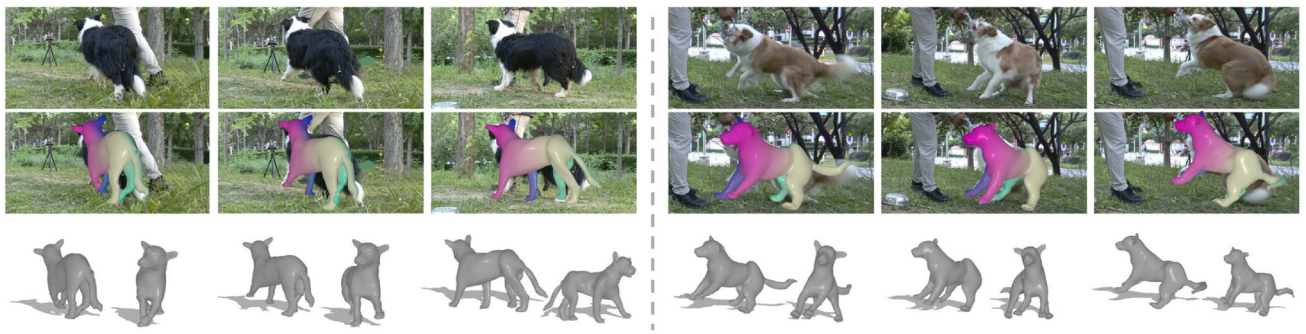
Fig. 6 Qualitative results. Each benchmark setting shows two results, with three frames per result from left to right. Three rows show the input image, CSE model projection, and recovered mesh, respectively. In multi-view settings, we show one input view for clarity and visualize the mesh from two angles



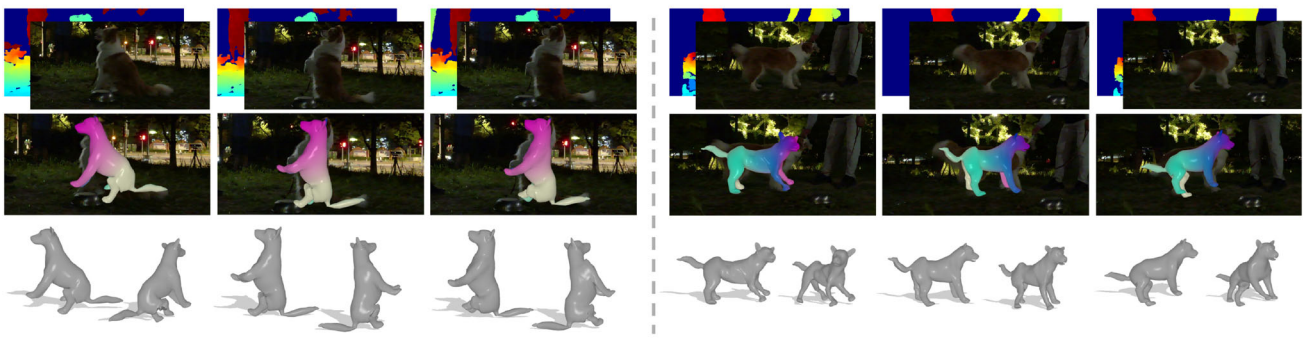
(a) Optimization Results on Single-view RGB Setting



(b) Optimization Results on Single-view RGB-D Setting

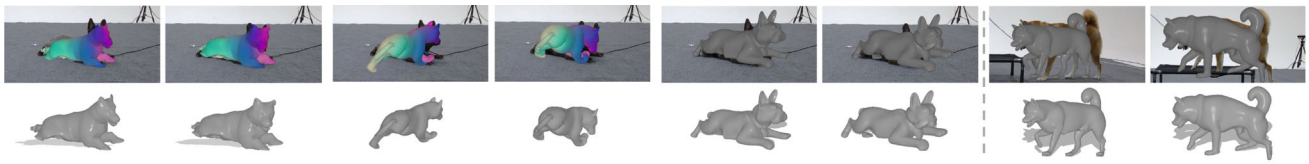


(c) Optimization Results on Multi-view RGB Setting

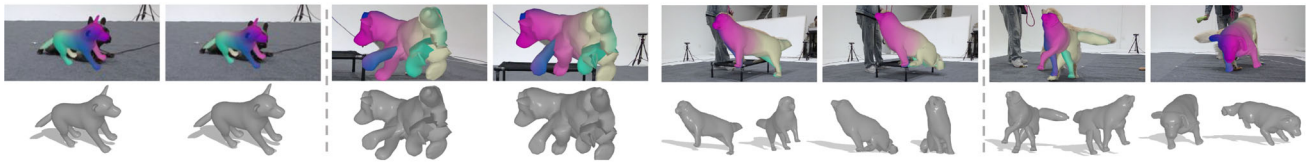


(d) Optimization Results on Multi-view RGB-D Setting

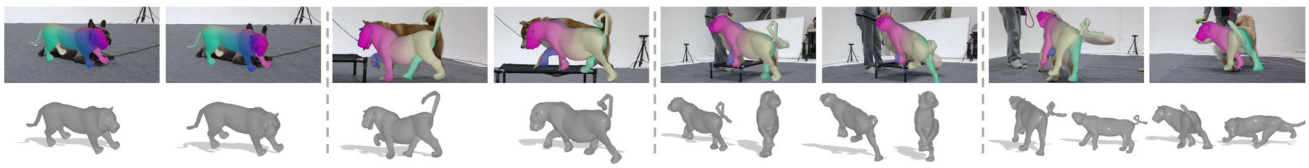
Fig. 7 Qualitative results on outdoor subset. Single-view RGB and multi-view RGB are evaluated on daytime recordings; single-view RGB-D and multi-view RGB-D are evaluated on nighttime recordings



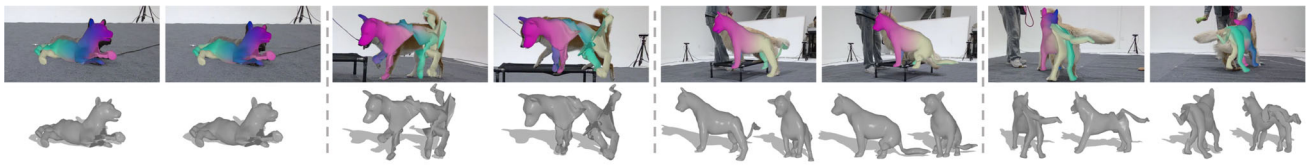
(a) Results of BITE (b) Results of SMALify (c) Results of SAM3D



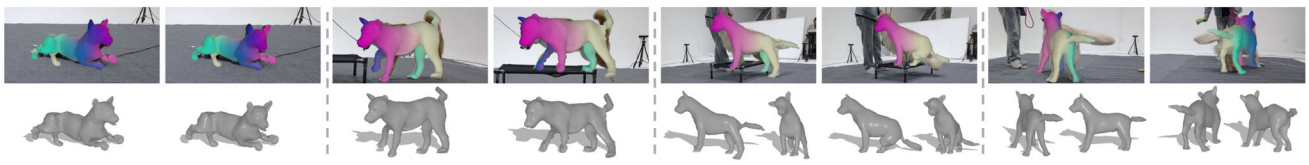
(d) Optimization Results of AnimalAvatar (e) Optimization Results of Multi-View SMAL-fit



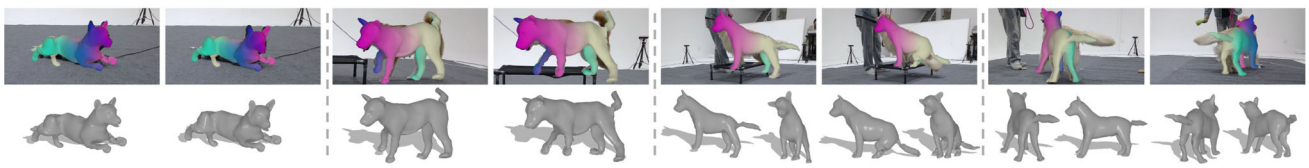
(f) Inference Results of AniMer



(g) Optimization Results of *w/o stage 1*

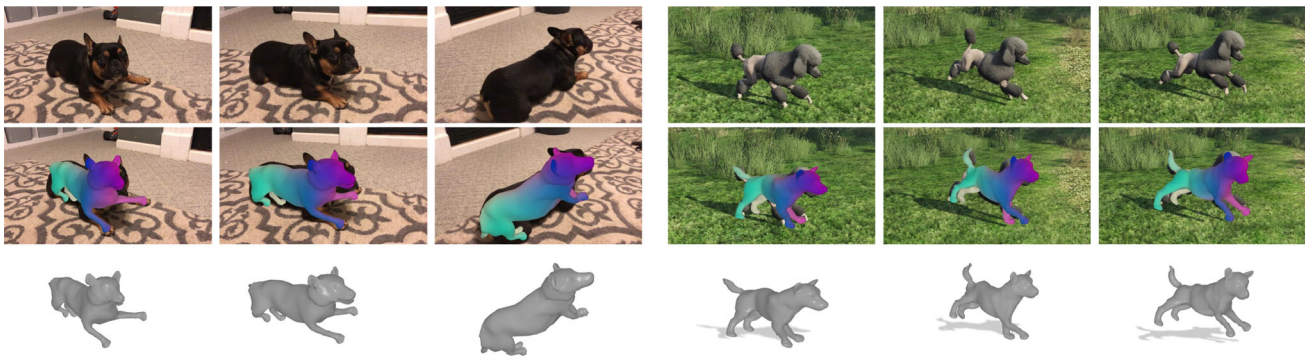


(h) Optimization Results of *w/o stage 3*



(i) Optimization Results of Our Full Method

Fig. 8 Qualitative result comparison. (**a**, **b**) show the single-view RGB recovery results of BITE and SMALify. **c** shows the qualitative results of SAM 3D under the single-view RGB and single-view RGB-D settings. **d** shows the optimization results of AnimalAvatar under the single-view RGB and single-view RGB-D settings. **e** shows the fitting results of the Multi-View SMAL-Fit baseline under the multi-view RGB and multi-view RGB-D settings. (**f**, **g** and **h**) show the results of the remaining baseline methods under four settings: single-view RGB, single-view RGB-D, multi-view RGB, and multi-view RGB-D, arranged from left to right. For each setting, we visualize two frames from the motion sequence



(a) Results on COP3D (b) Results on DigiDogs

Fig. 9 Generalization results. We further evaluate our method on (**a**) COP3D (Sinha et al., 2023) and (**b**) DigiDogs (Shooter et al., 2024a) datasets, which feature in-the-wild scenes

5.3 Additional Ablations

Simpler Temporal Smoothing Alternatives. We replace the smoothness optimization with simpler Kalman and spline smoothing in the third stage, which yields smoother results with reduced jitter. However, they also result in slightly lower IoU and F-score. Table 5 presents the results.

Impact of Different Initial Scaling Factors in w/o stage 1 Model and full Model As shown in Table 6, the *w/o stage 1* variant is highly sensitive to the initial scaling factor s. A small initialization ($s = 0.3$) tends to produce an undersized mesh that floats above the floor, leading to artificially low Pene$_{\%}$ and FS values because few valid contacts are detected. In contrast, a large initialization ($s = 1.0$) tends to produce an oversized mesh with severe floor penetration, resulting in a much higher Pene$_{\%}$. This indicates that, without Stage 1, the optimization can be strongly biased by the initial scale and converge to different local optima. By contrast, the *full* model is more robust to the choice of initial scale. The results under $s = 0.3$ and $s = 1.0$ are much closer across Pene$_{\%}$, Jitter, and FS, showing that Stage 1 effectively stabilizes the global scale and body placement before the subsequent joint optimization. This further supports the importance of the coarse alignment stage for robust motion recovery.

F-score at Different Thresholds. In the main quantitative comparison, we report F-score using a distance threshold of 0.05m as the default setting. To provide a more complete evaluation, we further report F-score under multiple thresholds, including 0.01m, 0.03m, 0.05m, 0.07m, 0.10m, and 0.15m. The results are summarized in Table 7. As expected, the F-score decreases when a smaller threshold is used, since the evaluation criterion becomes stricter. Nevertheless, our full model consistently outperforms the ablated variants across different thresholds. This indicates that the improvement of our method is not tied to a particular threshold choice, but remains stable under both strict and relatively relaxed distance criteria.

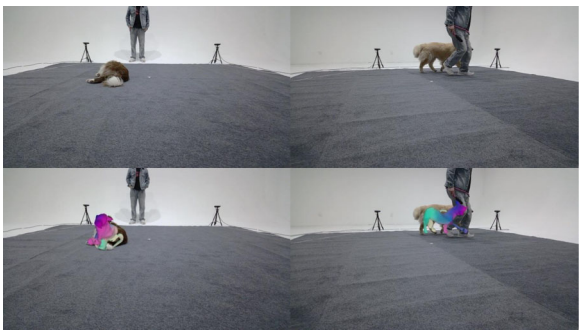


Fig. 10 Failure cases

6 Discussion

Limitations. While ***DogMo*** now includes a small complementary outdoor subset, its main benchmark is still collected in a controlled indoor lab scenario. The outdoor subset introduces additional visual variations, such as natural lighting changes, outdoor backgrounds, and low-light observations, but it remains limited in scale, subject diversity, and scene diversity. Therefore, learning-based models trained solely on ***DogMo*** may still face generalization challenges in more complex in-the-wild scenarios with diverse backgrounds, lighting conditions, and occlusions.

DogMo has subject-level imbalance and an uneven sequence-length distribution. These issues mainly result from the practical difficulty of collecting semantically controlled canine motions: different dogs vary in trainability and willingness to follow commands, and different action categories naturally have different durations. Although these distributions reflect realistic challenges in animal motion capture rather than arbitrary sampling or truncation, they may still affect learning-based methods trained on ***DogMo***.

Another limitation is the 15 FPS capture rate, which may be insufficient for fast motions such as jumping and running. Rapid movements can cause motion blur and temporal alias-

Table 5 Quantitative results with simpler temporal smoothing alternatives

Setting	Method	IoU ↑	IoU_{w5} ↑	F-score ↑	Jitter ↓
Single-view RGB	Kalman	0.6251	0.5015	0.3119	0.0211
	Spline	0.4821	0.2968	0.3426	0.0047
	Ours	0.6505	0.5247	0.3593	0.0249
Single-view RGB-D	Kalman	0.6248	0.4945	0.6889	0.0178
	Spline	0.4812	0.3031	0.6951	0.0037
	Ours	0.6414	0.5228	0.7673	0.0162
Multi-view RGB	Kalman	0.6021	0.3937	0.8586	0.0165
	Spline	0.4769	0.2379	0.8080	0.0036
	Ours	0.6258	0.4490	0.9107	0.0142
Multi-view RGB-D	Kalman	0.5907	0.3847	0.8595	0.0159
	Spline	0.4748	0.2337	0.8089	0.0045
	Ours	0.6286	0.4549	0.9137	0.0143

Table 6 Impact of different initial s in *w/o stage 1* model and *full* model

Method	$Pene_{\%}$ ↓	Jitter ↓	FS ↓	Method	$Pene_{\%}$ ↓	Jitter ↓	FS ↓
w/o stage 1 ($s = 0.3$)	0.0796	0.0056	0.0019	full ($s = 0.3$)	0.3516	0.0205	0.0327
w/o stage 1 ($s = 1.0$)	0.4077	0.0096	0.0147	full ($s = 1.0$)	0.3754	0.0233	0.0325

Table 7 F-score at different thresholds

Method	Single-view RGB						Single-view RGB-D					
	0.01	0.03	0.05	0.07	0.1	0.15	0.01	0.03	0.05	0.07	0.1	0.15
w/o leg-cross (Ours)	0.2127	0.2893	0.3575	0.3902	0.4338	0.4854	0.4394	0.5840	0.6619	0.6991	0.7387	0.7543
w/o scale (Ours)	0.0958	0.1366	0.1860	0.1862	0.2064	0.2294	0.3871	0.5443	0.6169	0.6244	0.6635	0.6808
w/o stage 1 (Ours)	0.1815	0.2384	0.3360	0.3695	0.3813	0.4143	0.4153	0.5495	0.6619	0.6860	0.6959	0.7305
w/o stage 3 (Ours)	0.2029	0.3027	0.3576	0.3915	0.4281	0.4722	0.4959	0.6801	0.7666	0.8131	0.8525	0.8833
Full (Ours)	0.2120	0.3086	0.3593	0.3952	0.4346	0.4789	0.5001	0.6863	0.7673	0.8139	0.8533	0.8841
Method	Multi-view RGB						Multi-view RGB-D					
	0.01	0.03	0.05	0.07	0.1	0.15	0.01	0.03	0.05	0.07	0.1	0.15
w/o leg-cross (Ours)	0.7917	0.8737	0.9086	0.9130	0.9168	0.9169	0.8116	0.8910	0.9078	0.9109	0.9193	0.9256
w/o scale (Ours)	0.8135	0.8838	0.9021	0.9034	0.9083	0.9091	0.8279	0.8838	0.9032	0.9073	0.9090	0.9123
w/o stage 1 (Ours)	0.8019	0.8715	0.9096	0.9137	0.9151	0.9167	0.8262	0.8843	0.9134	0.9156	0.9164	0.9181
w/o stage 3 (Ours)	0.8234	0.8959	0.9107	0.9148	0.9177	0.9196	0.8378	0.8846	0.9137	0.9172	0.9202	0.9212
Full (Ours)	0.8402	0.9006	0.9107	0.9148	0.9177	0.9196	0.8455	0.9036	0.9137	0.9178	0.9205	0.9222

ing, making masks, keypoints, and dense correspondences less reliable, and thus affecting motion recovery quality. These artifacts are common in low-cost real-world RGB-D capture, and remain practical challenges for future methods.

On the algorithmic side, although our optimization pipeline is effective, it remains computationally intensive, posing challenges for real-time or large-scale deployment.

Future Works. In future work, we aim to scale up ***DogMo*** to support more robust and generalizable modeling of dog motion. This expansion includes augmenting the dataset with photorealistic, in-the-wild backgrounds using advanced rendering and simulation tools. Beyond data augmentation, the expanded dataset will support two research directions: (1) incorporating physics-based constraints to enable physically plausible motion recovery, and (2) leveraging generative models, such as diffusion models, to learn data-driven motion priors for diverse motion synthesis (Yang et al., 2024; Wang et al., 2025) and prediction tasks.

7 Conclusion

This work introduces DogMo, a large-scale multi-view RGB-D dataset and benchmark for real canine motion recov-

ery, providing synchronized RGB-D observations and fitted SMAL-based 4D canine motion annotations. ***DogMo*** features 5 cameras capturing the movements of 10 dogs across 11 action types, resulting in $1.2k$ motion clips spanning over 220 minutes. Built upon this dataset, we establish four motion recovery benchmark settings to encourage future research. We also provide a practical three-stage fitting pipeline that recovers the dog's body parameters in a coarse-to-fine manner, from single-view to multi-view and from RGB to RGB-D inputs. We believe our work lays a strong foundation for advancing understanding of animal motion, with promising applications in animation and VR/AR.

Supplementary Information The supplementary material accompanying this article consists of a demonstration video that showcases the qualitative results and visual performance of the proposed method.

Appendix A Additional Details

A.1 Optimization

Optimization Rate. We set different optimization rates (*i.e.*, learning rates) for different parameters during the optimization. In the first stage, we initialize the optimization rates as follows: $5e^{-3}$ for the scale parameter and $5e^{-2}$ for the parameters of MLP_{TR}. In the second stage, we set the optimization rates to $5e^{-5}$ for the scale parameter, $5e^{-2}$ for the shape parameter, and $5e^{-4}$ for both MLP_{TR} and MLP_{θ}. In the third stage, we set the optimization rates to $1e^{-5}$ for both MLP_{TR} and MLP_{θ}. All optimization rates follow a step-wise decay schedule: at 75% and 93.75% of the total optimization steps, the rate is multiplied by a factor of 0.3.

Loss Weight. For each optimization loss term, we set the loss weight as follows: 400.0 for $\mathcal{L}_t^{\text{mask}}$, 60.0 for $\mathcal{L}_t^{\text{kp}}$, 1.0 for $\mathcal{L}_t^{\text{depth}}$, 20.0 for $\mathcal{L}_t^{\text{cse}}$, 2.5 for $\mathcal{L}_t^{\text{cross}}$, 0.005 for $\mathcal{L}_t^{\text{prior}}$, and 0.1 for $\mathcal{L}^{\text{temp}}$.

A.2 Shape & Pose Prior Loss

To enhance the plausibility of the estimated results and mitigate implausible joint poses or abnormal body proportions, we incorporate two prior constraint losses following existing studies (Rüegg et al., 2022; Zuffi et al., 2017): a pose prior loss and a shape prior loss. These priors are modeled as Gaussian distributions derived from the statistical properties of animal datasets. The Mahalanobis distance is employed during optimization to regularize the parameters, guiding the SMAL predictions toward anatomically consistent pose and body shape distributions.

Shape Prior Loss. The shape prior loss $\mathcal{L}^{\text{shape}}$ consists of two components: a body shape prior loss $\mathcal{L}^{\text{body}}$ and a limb proportion prior loss $\mathcal{L}^{\text{limb}}$. The body shape prior loss $\mathcal{L}^{\text{body}}$ is derived from a Gaussian prior fitted to the body shape parameters in the BARC (Rüegg et al., 2022) dataset. Here, $\boldsymbol{\mu}_{\text{body}}$ and $\boldsymbol{\Sigma}_{\text{body}}$ represent the mean vector and the positive semi-definite covariance matrix computed from the data. The Mahalanobis distance between the optimized shape vector $\boldsymbol{\beta}$ and the prior distribution is then given by

$$\mathcal{L}^{\text{body}} = (\boldsymbol{\beta} - \boldsymbol{\mu}_{\text{body}})^{\top} \boldsymbol{\Sigma}_{\text{body}}^{-1} (\boldsymbol{\beta} - \boldsymbol{\mu}_{\text{body}}). \tag{A1}$$

To further ensure reasonable limb proportions, we introduce a limb proportion prior loss $\mathcal{L}^{\text{limb}}$, which applies a weighted quadratic penalty to limb-related shape coefficients, *i.e.*,

$$\mathcal{L}^{\text{limb}} = \left\| \mathbf{w} \odot \boldsymbol{\beta}^{\text{limb}} \right\|_2^2, \tag{A2}$$

where $\mathbf{w}$ is a manually defined weight vector. The final shape prior loss combines the body and limb prior terms through a weighted sum, *i.e.*, $\mathcal{L}^{\text{shape}} = w_{\text{body}} \cdot \mathcal{L}^{\text{body}} + w_{\text{limb}} \cdot \mathcal{L}^{\text{limb}}$.

Pose Prior Loss. Similarly, we model the joint rotation prior as a Gaussian distribution characterized by the mean vector $\boldsymbol{\mu}_{\text{pose}}$ and a positive semi-definite covariance matrix $\boldsymbol{\Sigma}_{\text{pose}}$. The Mahalanobis distance between the optimized pose vector and this prior distribution is then computed as

$$\mathcal{L}_t^{\text{pose}} = (\theta_t - \boldsymbol{\mu}_{\text{pose}})^{\top} \boldsymbol{\Sigma}_{\text{pose}}^{-1} (\theta_t - \boldsymbol{\mu}_{\text{pose}}). \tag{A3}$$

A.3 Metric Computation

F-score. Following Wang et al. (2018), we first uniformly sample points from our optimization result and lift the masked depth image pixels into 3D space. We calculate precision and recall by measuring the percentage of points from the optimization or the depth image reprojection that have a nearest neighbor in the other set within a specified threshold. We set the threshold as 0.05. The F-score is then computed as the harmonic mean of precision and recall.

Pene$_{\%}$. We compute the penetration ratio as the proportion of SMAL mesh vertices that penetrate the ground plane in each frame.

Jitter. The jitter metric quantifies the temporal smoothness of predicted SMAL joint trajectories, indicating the stability of the motion. It is calculated as the mean second-order difference of all joint positions across the sequence.

Foot Skating. Foot skating measures the unintended horizontal displacement of the feet while in contact with the ground, indicating unrealistic gait motion. The metric is calculated by averaging the horizontal velocity of the feet whenever their distance to the ground plane is below 0.04m.

A.4 Camera Calibration and Synchronization

We calibrate the five ORBBEC Astra 2 RGB-D cameras before data collection. Each camera is first calibrated to obtain its intrinsic parameters, and adjacent camera pairs are then calibrated to estimate relative poses. All cameras are registered into a shared world coordinate system. The average reprojection error is approximately 0.7 pixels.

For synchronization, we use the hardware synchronization setup of the ORBBEC cameras. The temporal offset between adjacent cameras is 4 ms, resulting in a maximum offset of 16 ms between the first and fifth cameras.

A.5 Annotation

Masks. We use Grounded SAM 2 (Ren et al., 2024) with the text prompt "dog" to generate dog masks for RGB images from all five views. ***DogMo*** contains approximately $\sim$ 1M mask annotations. The generated masks are manually checked to reduce errors caused by occlusion, background clutter, or segmentation failures.

Keypoints. We use the dog keypoint detector from BARC (Rüegg et al., 2022) to annotate 2D keypoints for all five views. The detector provides keypoint confidence scores but not binary visibility flags. We discard keypoints with confidence scores lower than 0.2 and manually verify the generated annotations.

Action labels. Each motion clip is assigned one semantic action label according to the dominant behavior in the valid motion segment. Since dogs cannot always follow instructions consistently, the number of valid clips varies across subjects and action categories.

A.6 Implementation Details of Baselines and Ablations

Except for AnimalAvatar (Sabathier et al., 2024), the compared methods are originally designed for single-frame pose and shape estimation. We therefore apply them frame by frame to each video and concatenate the per-frame results in temporal order.

For BITE Rüegg et al., (2023), we use the official pretrained model and evaluate it under the single-view RGB setting. For AniMer (Lyu et al., 2025), we train the model on ***DogMo*** using 2D weak supervision from masks and keypoints, with a random 70%/30% train/test split. In the multi-view setting, synchronized images are jointly used as input, and the estimated mesh is projected back to all input views. We compute the losses in each view and sum them for optimization. Depth supervision is additionally used in RGB-D settings. For SMALify (Biggs, 2020; Biggs et al., 2020), we use the public implementation and default configuration. For AnimalAvatar (Sabathier et al., 2024), we use the default configuration and adapt it to RGB-D inputs by adding depth supervision.

For the multi-view settings, we also implement Multi-View SMAL-Fit. We triangulate 2D keypoints from all views into 3D keypoints using calibrated camera parameters, and then fit SMAL by minimizing joint errors, together with silhouette/depth alignment terms when available and standard shape/pose priors.

We additionally include SAM 3D (Chen et al., 2026) as a qualitative baseline only, since it reconstructs a single-frame mesh but does not recover metric scale, articulated skeletons, or temporally consistent motion.

We evaluate four ablated variants: w/o stage 1, w/o stage 3, w/o leg-cross, and w/o scale, corresponding to removing coarse alignment, temporal refinement, leg-cross regularization, and the global scaling factor s, respectively.

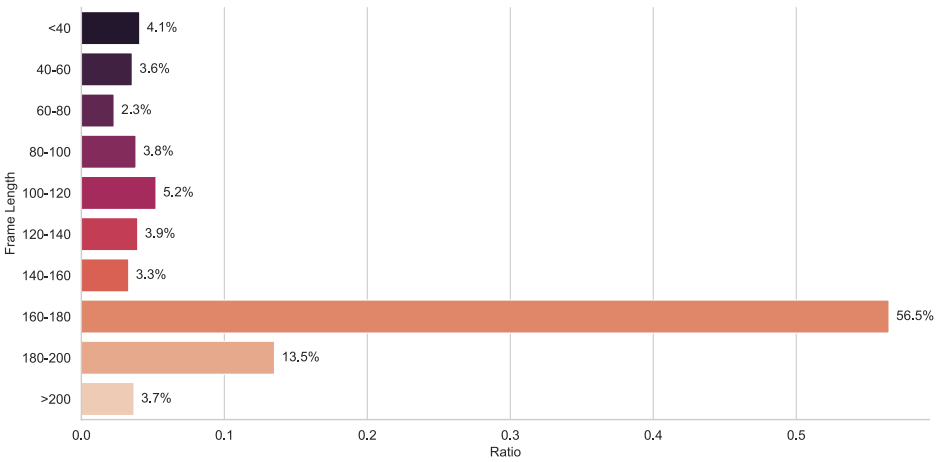
A.7 Floor Plane Estimation

For Pene% and Foot Skating, we estimate a floor plane from depth observations. After camera calibration, we select several floor pixels from one depth frame, back-project them into 3D world coordinates using camera intrinsics, extrinsics, and depth values, and fit a 3D plane to these points. The fitted plane is used as the floor for computing physical plausibility metrics.

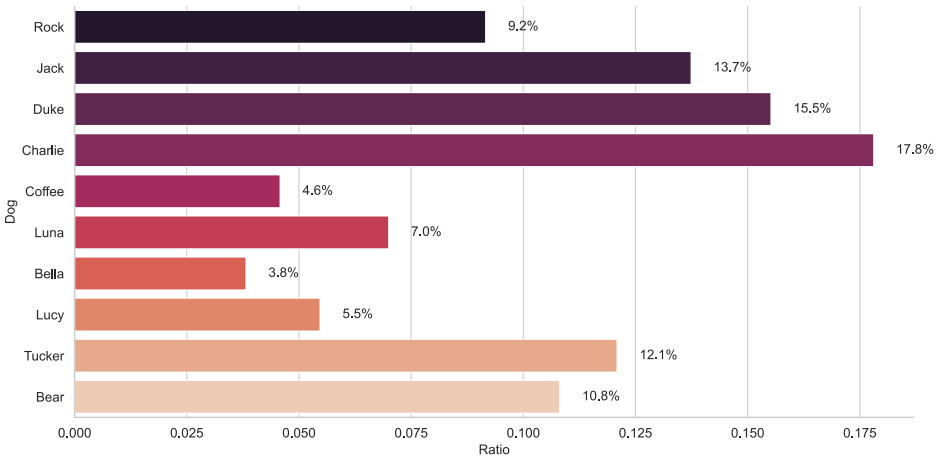
Appendix B Dataset Statistics

Statistics. We provide additional statistics for the ***DogMo*** dataset in Fig. 11. Figure 11a illustrates the distribution of motion sequence lengths. In ***DogMo***, frame counts range from 13 to 286, with over half of the sequences falling between 180 and 200 frames. The average frame length is approximately 155. This distribution is related to the natural duration of different action categories and our valid-clip extraction protocol. Short actions, such as running along a straight path or jumping over an obstacle, often last only a brief period and thus produce shorter clips. In contrast, continuous or interactive actions, such as walking, playing with humans, or playing with toys, can be guided by the trainer for several seconds, typically around 5–7 seconds. Since these longer actions account for a large portion of the dataset, many clips fall into the 180–200 frame range. This distribution is therefore caused by action-dependent duration differences rather than arbitrary length truncation.

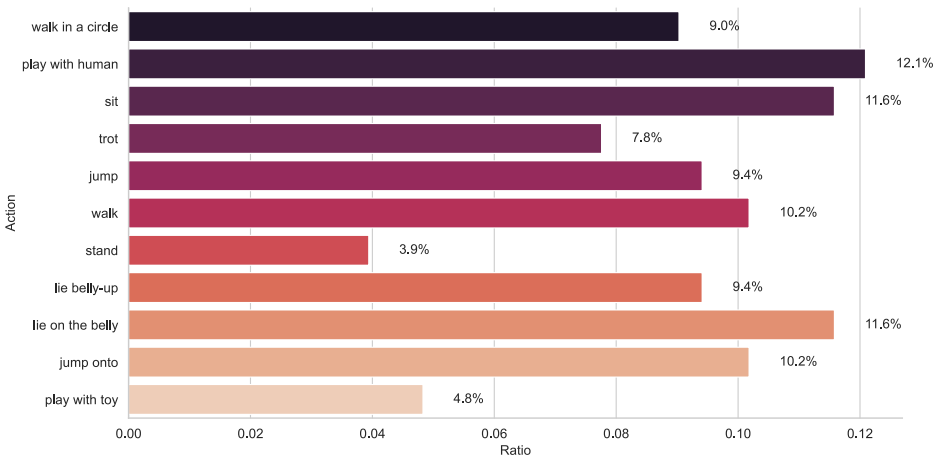
Figure 11b presents the distribution of motion data across 10 individual dog instances in ***DogMo***. The subject-level imbalance mainly stems from the practical difficulty of collecting semantically controlled animal motions. Unlike human subjects, dogs cannot be instructed to perform arbitrary actions consistently, and their ability or willingness to



(a) Frame Length Distribution



(b) Dog Instance Frequency



(c) Action Frequency

Fig. 11 Statistics of ***DogMo***. The action "stand" denotes the dog standing upright on its hind legs

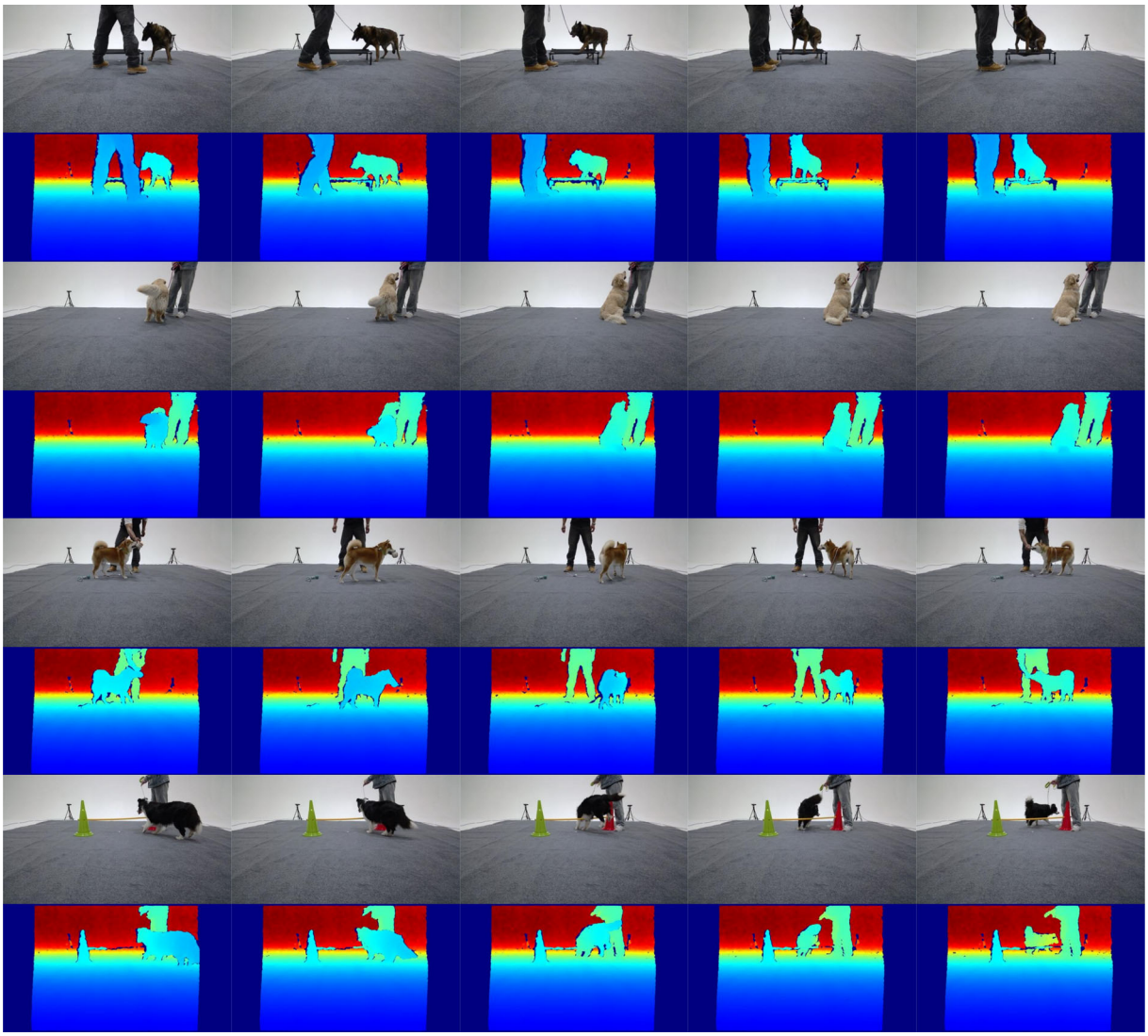


Fig. 12 Motion recording visualization. We visualize four motion recordings, each presenting both an RGB sequence and a depth sequence

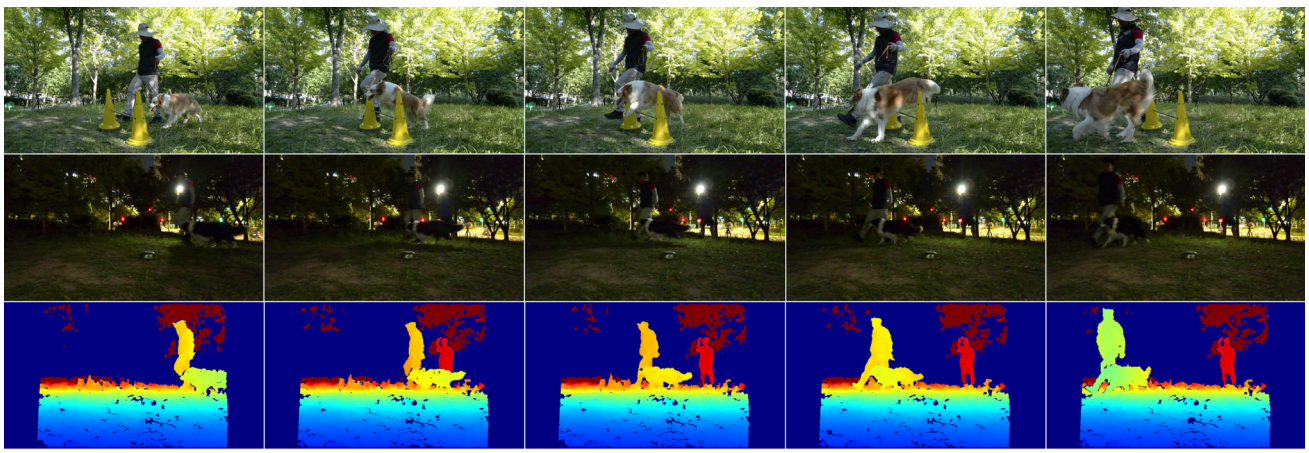


Fig. 13 Motion recording visualization of outdoor subset. We visualize representative outdoor motion-capture recordings from both daytime and nighttime conditions. Since the infrared-based depth sensing of the ORBBEC Astra 2 becomes unreliable outdoors due to sunlight interference, we only show RGB images for the daytime recording

follow commands varies across individuals. For example, some dogs, *e.g.*, Charlie, are more responsive to the trainers, while actions such as standing on hind legs or certain human-dog interactions can only be reliably performed by a subset of dogs. Therefore, some dog instances naturally contribute more valid clips than others. This imbalance reflects the realistic challenges of collecting labeled canine motion data, and may influence learning-based methods trained on ***DogMo***.

Additionally, each motion sequence in ***DogMo*** is annotated with one of 11 predefined action categories. The motion distribution over these action types is shown in Fig. 11c.

Visualization of Motion Recordings. We additionally provide visualizations of the motion recordings in Fig. 12 and Fig. 13. For video demonstrations, please refer to the supplementary demo video.

Appendix C Animal Welfare and Capture Protocol

We provide additional details about the animal welfare considerations during data collection. All dog motion recordings were conducted under the supervision of experienced dog trainers or dog owners. The dogs were guided to perform natural behaviors using verbal commands, gestures, toys, or food rewards, rather than physical force. The recorded actions, such as walking, running, sitting, jumping, playing with toys, and interacting with humans, are common daily behaviors for dogs.

During capture, the dogs were allowed to move freely within the capture area. No invasive devices, motion-capture markers, wearable sensors, or body-mounted equipment were attached to the dogs. The RGB-D cameras were placed around the capture area and recorded the dogs from a distance, so the capture process did not require direct physical contact with the animals.

We also allowed sufficient rest during the recording sessions. If a dog showed signs of fatigue, stress, distraction, or unwillingness to continue, the recording was paused or stopped. Since different dogs have different physical abilities, trainability, and willingness to follow commands, we did not force every dog to perform every action category. This also partially explains the subject-level and action-level imbalance in the dataset.

For the outdoor subset, the same welfare protocol was followed. The dogs were recorded in a safe outdoor environment under trainer or owner supervision. The recording area was selected to avoid dangerous obstacles, traffic, or other potential risks. Nighttime recordings were conducted under controlled lighting conditions to ensure that the dogs could move safely.

Overall, the data collection process was designed to capture natural canine behaviors while minimizing discomfort or risk to the animals.

Acknowledgements This project is supported by Beijing Natural Science Foundation (L242094), the State Key Laboratory of General Artificial Intelligence (SKLAGI2024OP15), and the National Natural Science Foundation of China (NSFC) (62172043).

Author Contributions Zan Wang designed the overall model, organized the experiments, and wrote the manuscript. Siyu Chen and Luya Mo implemented the model, conducted the experiments, contributed to data collection, and assisted in preparing several figures. Xinfeng Gao collected the dataset. Yuxin Shen contributed to part of the figure preparation. Lebin Ding participated in part of the data collection process. Wei Liang, as the supervising professor and corresponding author, oversaw the project and guided the research progress.

Funding This project is supported by Beijing Natural Science Foundation (L242094), the State Key Laboratory of General Artificial Intelligence (SKLAGI2024OP15), and the National Natural Science Foundation of China (NSFC) (62172043).

Availability of data and material The motion recordings in ***DogMo*** are available at HuggingFace repository. Besides, this study also uses publicly available datasets COP3D (Sinha et al., 2023) and DigiDogs (Shooter et al., 2024a), which are available at github.com and cvssp.org.

Declarations

Conflict of interest All authors certify that they have no conflict of interest or conflict of interest related to this work. All authors are affiliated with the Beijing Institute of Technology, and Wei Liang is additionally affiliated with the Yangtze Delta Region Academy of Beijing Institute of Technology, Jiaxing. None of the authors has any financial relationships, such as employment, consultancies, stock ownership, honoraria, paid expert testimony, or patent applications, that could be perceived as influencing the results or interpretation of this manuscript. The authors also declare that they have no non-financial conflict of interest, including personal, professional, or institutional relationships, that could be viewed as influencing the research.

Ethics approval and consent to participate This study does not involve human subjects or physiological experiments. The dataset in this work is collected from videos of dogs in natural environments without invasive procedures or interventions. According to our institution's research ethics policy, IRB (Institutional Review Board) approval is only required for studies involving human participants or biological experimentation. Therefore, no IRB approval is necessary or applicable for this research.

Consent for publication Not applicable. No identifiable individual data is included in this work.

Materials availability This study involved physical materials including the ORBBEC Astra 2 RGB-D cameras, as illustrated in the manuscript. The data used in this work were collected by the authors and the availability is described above.

Code availability The code for DogMo is publicly available at https://github.com/BIT-PIE/DogMo.

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